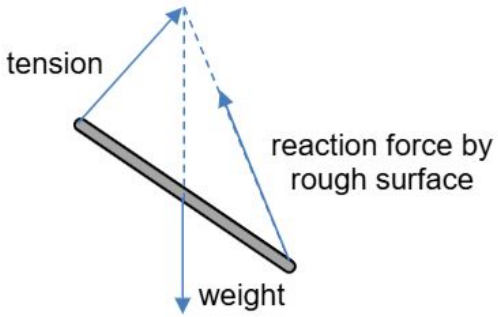
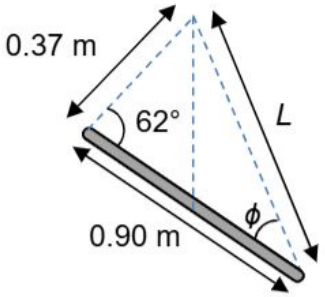
Qn			Answers	Marks
1	(a)		The principle of moments states that if a body is in equilibrium, the sum of all the clockwise moments about any axis/point must be equal to the sum of all the anticlockwise moments about the same axis/point.	B1
	(b)		 To ensure that the lines of action approximately pass through same point.	B1
	(c)		 Recognise that the point of intersection of all 3 forces meet at the fixed point. (rule of concurrency) Let L be the distance from the fixed point to the bottom end of rod Use cosine rule to find distance to top $L^2 = (0.90)^2 + (0.37)^2 - 2(0.90)(0.37)\cos(62^\circ)$ $L = 0.796271$ m	B1
			Use sine rule to find angle ϕ $\frac{\sin\phi}{0.37} = \frac{\sin 62^\circ}{0.796271}$ $\phi = 24.3^\circ$ ∴ angle by ground on rod = 40° + 24.3° = 64.3° (shown)	B1
	(d)		By taking moments about the upper end of rod, (0.200)(9.81)(0.45) cos 40° = F_{ground} (0.90)(sin 24.3°) F_{ground} = 1.82 N	M1 A1

	(e)		All four forces will have their lines of action still meeting at the fixed point that the string is attached to.	B1
			Applying the principle of moments to take moments about that point, the reaction force by the rough surface must continue to point at the fixed point, such that there is no net moment about that point. Thus, angle in **(c)** would remain the same.	B1
	(f)		If the horizontal surface were frictionless, the surface can only exert a force that is vertically upward. Since weight also acts vertically downward, the vector sum of these 2 forces do not have a horizontal component.	M1
			Thus, the force from the string must also have no horizontal component and the string would be vertical. The angle between the string and rod must increase.	A1

Qn			Answers	Marks